

Task 3: Data Analysis with Pandas

```
import pandas as pd

import os

print("\n=====")

print("    DATA ANALYSIS WITH PANDAS PROJECT")

print("=====\\n")

# -----

# STEP 1 : TAKE CSV FILE INPUT FROM USER

# -----

file_path = input("Enter CSV file path: ")

# Check whether file exists

if not os.path.exists(file_path):

    print("\\nERROR : File does not exist.")

    exit()

try:

    # Load CSV dataset

    df = pd.read_csv(file_path)

    print("\\n=====")

    print("DATASET LOADED SUCCESSFULLY")

    print("=====\\n")

    # -----

    # STEP 2 : INSPECT DATASET

    # -----

    print("FIRST 5 ROWS OF DATASET:\\n")

    print(df.head())

    print("\\n=====")

    print("\\nDATASET INFORMATION:\\n")

    print(df.info())

    print("\\n=====")

    print("\\nSTATISTICAL SUMMARY:\\n")

    print(df.describe(include='all'))

    print("\\n=====")

    # -----

    # STEP 3 : CHECK MISSING VALUES

    # -----

    print("\\nMISSING VALUES IN EACH COLUMN:\\n")

    print(df.isnull().sum())
```

```

print("\n=====")
# -----
# STEP 4 : CLEAN MISSING DATA
# -----
print("\nCLEANING MISSING DATA...\n")
# Fill missing numeric values with mean
numeric_columns = df.select_dtypes(include=['number']).columns
for col in numeric_columns:
    mean_value = df[col].mean()
    df[col].fillna(mean_value, inplace=True)
# Fill missing object/string values with "Unknown"
object_columns = df.select_dtypes(include=['object']).columns
for col in object_columns:
    df[col].fillna("Unknown", inplace=True)
print("Missing values cleaned successfully.")
print("\nUPDATED MISSING VALUES:\n")
print(df.isnull().sum())
print("\n=====")
# -----
# STEP 5 : REMOVE DUPLICATE ROWS
# -----
duplicate_count = df.duplicated().sum()
print(f"\nTOTAL DUPLICATE ROWS FOUND: {duplicate_count}")
df.drop_duplicates(inplace=True)
print("Duplicate rows removed successfully.")
print("\n=====")
# -----
# STEP 6 : DISPLAY COLUMN NAMES
# -----
print("\nAVAILABLE COLUMNS:\n")
for col in df.columns:
    print("-", col)
print("\n=====")
# -----
# STEP 7 : FILTERING DATA
# -----
choice = input(

```

```

"\nDo you want to filter data? (yes/no): "

).lower()

if choice == "yes":

    filter_column = input("Enter column name for filtering: ")

    if filter_column in df.columns:

        filter_value = input(

            "Enter value to filter rows: "

        )

        filtered_data = df[

            df[filter_column].astype(str) == filter_value

        ]

        print("\nFILTERED DATA:\n")

        print(filtered_data)

    else:

        print("\nInvalid column name.")

        print("\n=====")

# -----

# STEP 8 : GROUPING AND AGGREGATION

# -----

group_choice = input(

    "\nDo you want grouping and aggregation? (yes/no): "

).lower()

if group_choice == "yes":

    group_column = input(

        "Enter column name for grouping: "

    )

    if group_column in df.columns:

        # Select numeric columns only

        numeric_cols = df.select_dtypes(include='number').columns

        if len(numeric_cols) > 0:

            grouped_data = df.groupby(group_column)[numeric_cols].mean()

            print("\nGROUPED DATA WITH MEAN VALUES:\n")

            print(grouped_data)

        else:

            print("\nNo numeric columns available.")

    else:

        print("\nInvalid column name.")

```

```

print("\n=====")
# -----
# STEP 9 : SIMPLE INSIGHTS
# -----

print("\nINSIGHT SUMMARY")

print("=====\n")

print(f"Total Rows   : {df.shape[0]}")

print(f"Total Columns : {df.shape[1]}")

print("\nColumn Names:")

print(list(df.columns))

# Numeric insights

numeric_cols = df.select_dtypes(include='number').columns

if len(numeric_cols) > 0:

    print("\nNUMERIC COLUMN INSIGHTS:\n")

    for col in numeric_cols:

        print(f"Column Name : {col}")

        print(f"Average   : {df[col].mean():.2f}")

        print(f"Maximum    : {df[col].max()}")

        print(f"Minimum    : {df[col].min()}")

        print("-----")

    # Object/String insights

    object_cols = df.select_dtypes(include='object').columns

    if len(object_cols) > 0:

        print("\nCATEGORICAL COLUMN INSIGHTS:\n")

        for col in object_cols:

            print(f"Most Common Value in '{col}':")

            print(df[col].mode()[0])

            print("-----")

        print("\n=====")

    print("DATA ANALYSIS COMPLETED SUCCESSFULLY")

    print("=====\n")

except FileNotFoundError:

    print("\nERROR : File not found.")

except pd.errors.EmptyDataError:

    print("\nERROR : CSV file is empty.")

except pd.errors.ParserError:

    print("\nERROR : Problem while parsing CSV file.")

```

except Exception as e:

print("\nUNEXPECTED ERROR OCCURRED:")

print(e)

```
=====
DATA ANALYSIS WITH PANDAS PROJECT
=====
Enter CSV file path: Example Output (Sample Run)
```text
=====
DATA ANALYSIS WITH PANDAS PROJECT
=====
Enter CSV file path: student_data.csv

DATASET LOADED SUCCESSFULLY

=====
FIRST 5 ROWS OF DATASET:

 ID Name Age Department Marks
0 1 Amit 20 CS 85
1 2 Rahul 21 IT 78
2 3 Sneha 20 CS 92
3 4 Priya 22 EE 88
4 5 Karan 21 IT 76

=====
DATASET INFORMATION:

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 5 columns):
Column Non-Null Count Dtype
--- -
0 ID 5 non-null int64
1 Name 5 non-null object
2 Age 5 non-null int64
3 Departm 5 non-null object
4 Marks 5 non-null int64
ERROR : File does not exist.

=====
STATISTICAL SUMMARY:

 ID Age Marks
count 5.000000 5.0 5.000000
mean 3.000000 20.8 83.800000
std 1.581139 0.8 6.610598
min 1.000000 20.0 76.000000
25% 2.000000 20.0 78.000000
50% 3.000000 21.0 85.000000
75% 4.000000 21.0 88.000000
max 5.000000 22.0 92.000000
=====
```

MISSING VALUES IN EACH COLUMN:

ID	0
Name	0
Age	0
Department	0
Marks	0

CLEANING MISSING DATA...

Missing values cleaned successfully.

UPDATED MISSING VALUES:

ID	0
Name	0
Age	0
Department	0
Marks	0

TOTAL DUPLICATE ROWS FOUND: 0

Duplicate rows removed successfully.

AVAILABLE COLUMNS:

- ID
- Name
- Age
- Department
- Marks

Do you want to filter data? (yes/no): yes

Enter column name for filtering: Department

Enter value to filter rows: CS

FILTERED DATA:

	ID	Name	Age	Department	Marks
0	1	Amit	20	CS	85
2	3	Sneha	20	CS	92

Do you want grouping and aggregation? (yes/no): yes

Enter column name for grouping: Department

GROUPED DATA WITH MEAN VALUES:

	ID	Age	Marks
Department			
CS	2.0	20.0	88.5
EE	4.0	22.0	88.0
IT	3.5	21.0	77.0

=====

INSIGHT SUMMARY

=====

Total Rows : 5

Total Columns : 5

Column Names:

['ID', 'Name', 'Age', 'Department', 'Marks']

NUMERIC COLUMN INSIGHTS:

Column Name : ID

Average : 3.00

Maximum : 5

Minimum : 1

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Column Name : Age

Average : 20.80

Maximum : 22

Minimum : 20

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Column Name : Marks

Average : 83.80

Maximum : 92

Minimum : 76

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CATEGORICAL COLUMN INSIGHTS:

Most Common Value in 'Name':

Amit

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Most Common Value in 'Department':

CS

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DATA ANALYSIS COMPLETED SUCCESSFULLY

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...Program finished with exit code 0

Press ENTER to exit console.